INTRODUCTION TO DATABASE SYSTEMS



From fundamentals to the AI era

Disclaimer: The images in these slides are AI-generated

Learning objectives

After this lecture, students will be able to:

- Understand what a database is and why databases are needed
- Understand core concepts: Table, Row, Column, Primary Key
- Know the role of a DBMS
- Have an initial idea of what SQL is used for
- Know the history and the trends of database research

The invisible backbone of modern apps



Database infrastructure

A database is a collection of structured data, engineered for fast searching, safe update, and concurrent access

The limits of ordinary files

Ordinary files



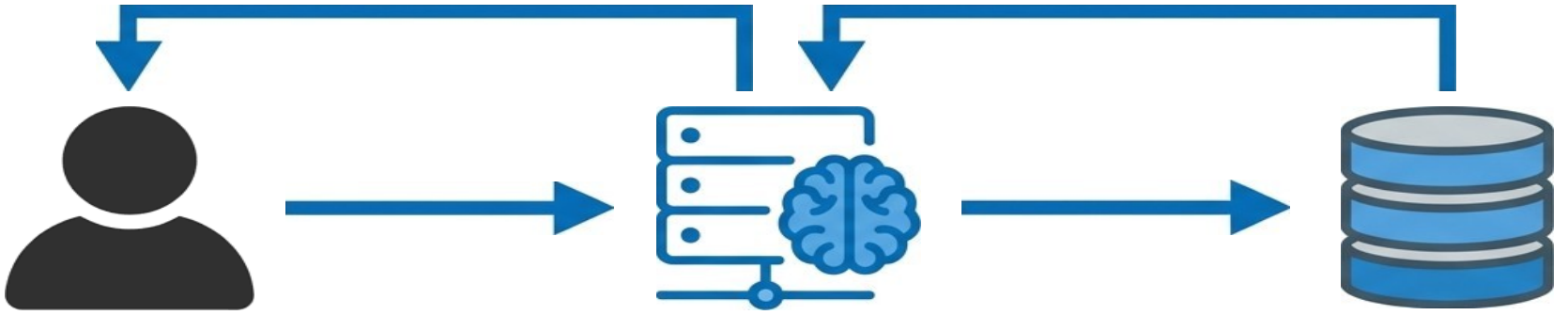
- High data redundancy
- Single-user access only
- Weak security
- Easy to break

Database system



- Reduced redundancy
- Concurrent access
- Built-in security
- Data integrity constraints

The brain vs. The memory



User

DBMS (The software)

- MySQL
- Oracle
- PostgreSQL
- SQL Server

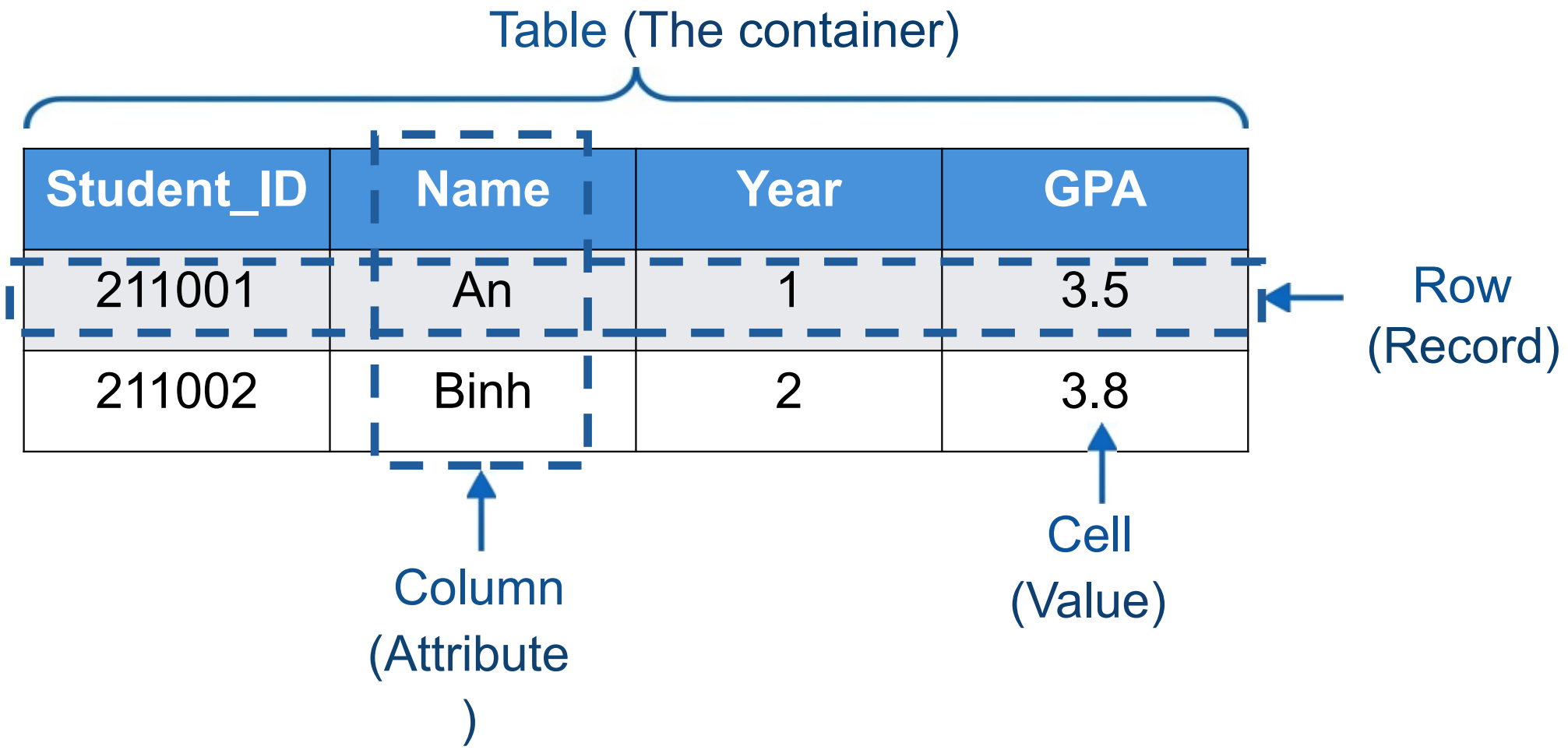
Database
(The data)

The Client-Server model: Applications never touch the data files directly; they ask the DBMS.

Applications of DBMS



Anatomy of a table



Law and Order: Keys and Relationships

Primary Key



Unique Identifier for every row.
Rule: Must be unique,
cannot be null.

@Good: Student ID.

@Bad: Email (Changeable),
Name (Duplicate).

STUDENTS

Student_ID	Name	Year
211001	An	1
211002	Bob	2
211003	Colin	3
211004	Daisy	4

COURSES

Course_ID	Course_name	Student_ID
CS101	Intro to CS	211002
MA101	Calculus I	211001
CS102	Data structures	211003
CS102	Data structures	211001

One-to-many relationship

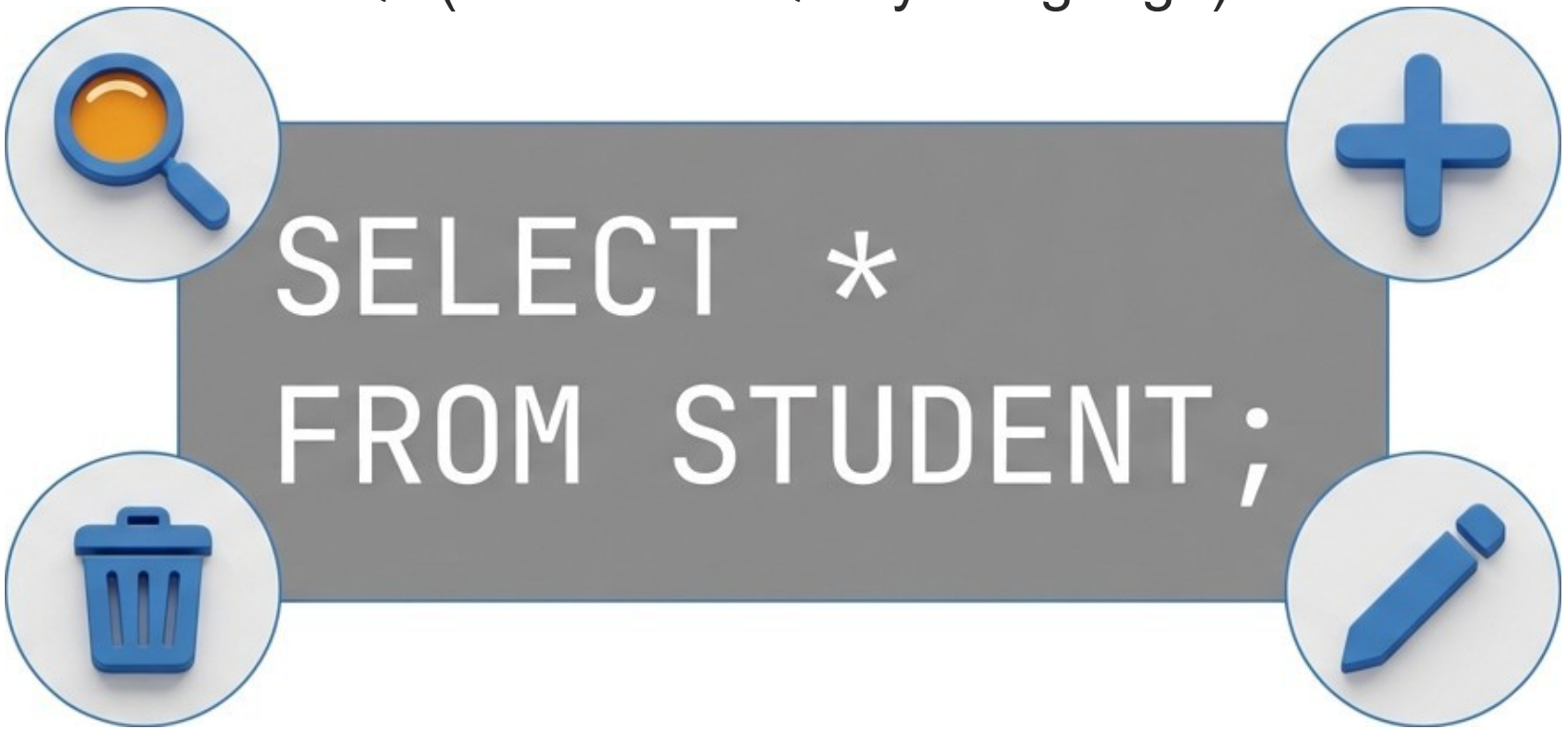
Tables connect via Keys

Speaking the language of data

SQL (Structured Query Language)

Retrieve (SELECT)

Insert (INSERT)



Delete (DELETE)

Update (UPDATE)

SQL is a data manipulation language, not a general programming language.

More about SQL

Learning the basics:

- https://www.w3schools.com/sql/
- https://sqlbolt.com/lesson/select_queries_introduction
- https://sqliteonline.com/

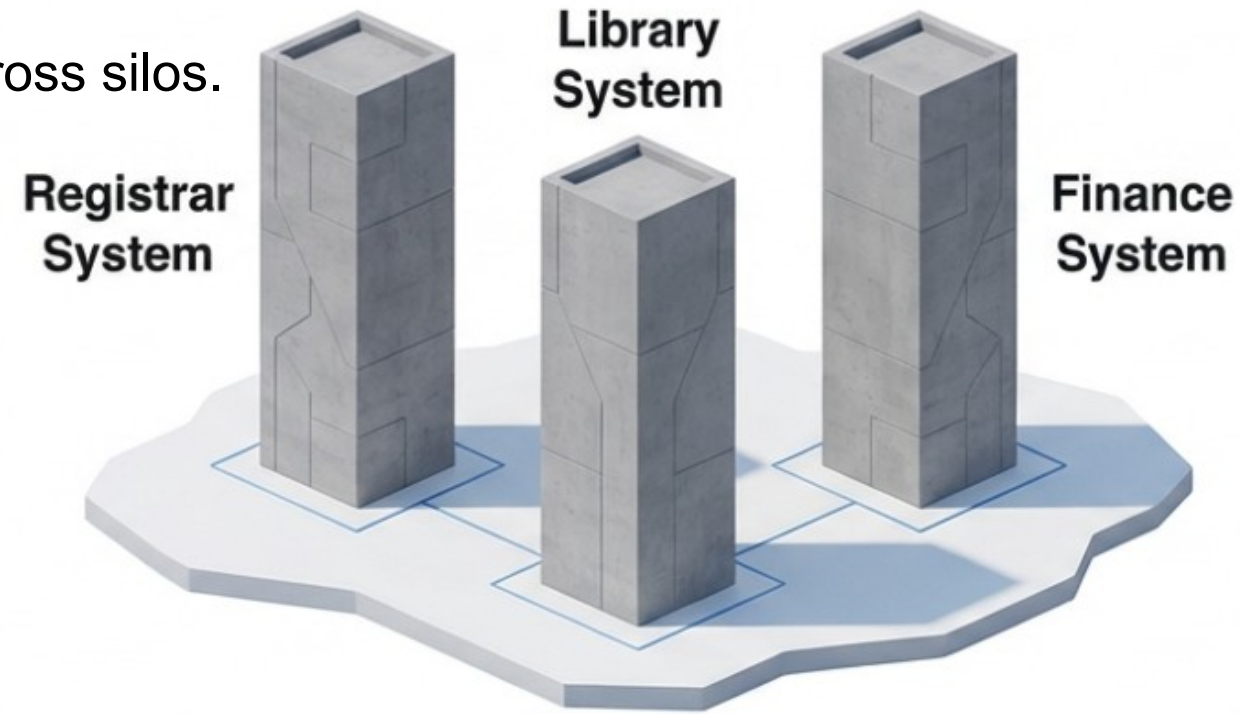
Advanced:

- https://leetcode.com/problems/combine-two-tables/description/
- https://www.hackerrank.com/domains/sql

1960s: The era of isolated files

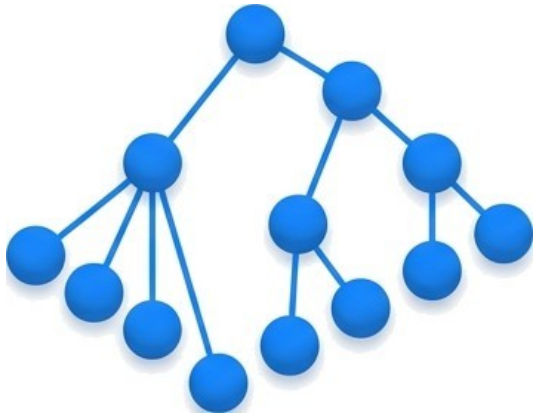
Pain points

- **Data redundancy:** Information repeated in each silo.
- **Inconsistency:** Updates don't sync across silos.
- **Rigidity:** Hard to modify.
- **No concurrency:** One user at a time.



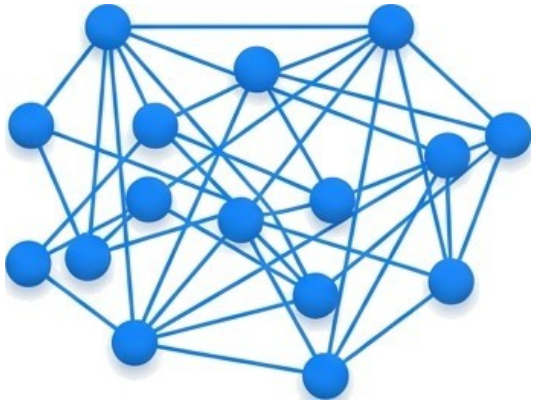
1970s: The struggle for structure

Hierarchical model



Data organized in a rigid tree structure.
Strict parent-child relationship.
Limited flexibility.

Network model



Data nodes connected in a flexible web.
Complex, many-to-many relationships.
Challenging navigation.

The fatal flaw: Navigational complexity

Programmers required knowledge of physical pointers to find data.
Changing storage broke the code.

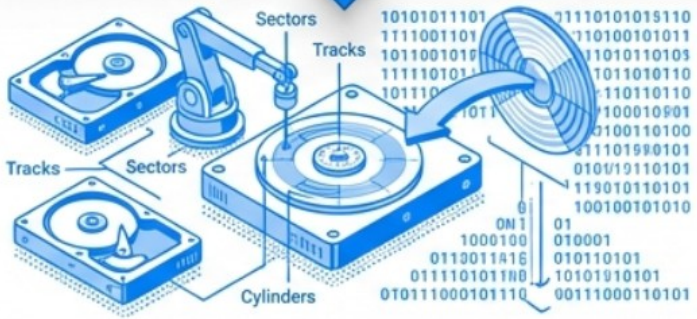
1980s: The relational revolution

Logical View
(What user sees)

ID	Name	Department	Salary
101	Alice	Sales	$60k
102	Mart	Department	$70k
103	Cosign	Tech	$70k
104	Kerry	Sales	$60k

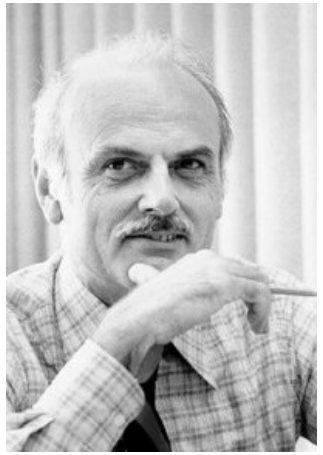
DBMS Abstraction

Physical Storage
(How it is stored)



A Relational Model of Data for Large Shared Data Banks

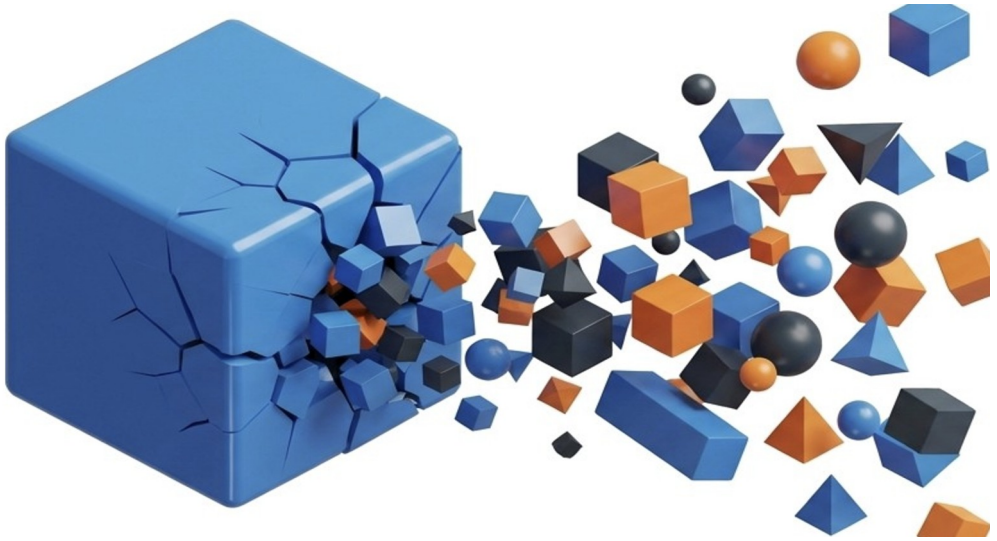
E. F. Codd
IBM Research Laboratory, San Jose, California

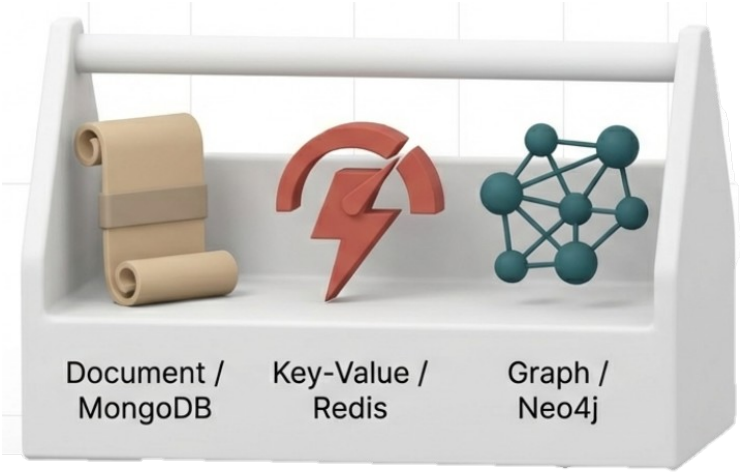


Foundation of

- MySQL
- PostgreSQl
- Oracle

2010s: The NoSQL pivot





The context:
Big data & Social media

The solution: NoSQL
(Not only SQL)

Schema flexibility:
Handle **unstructured data** without predefined tables.

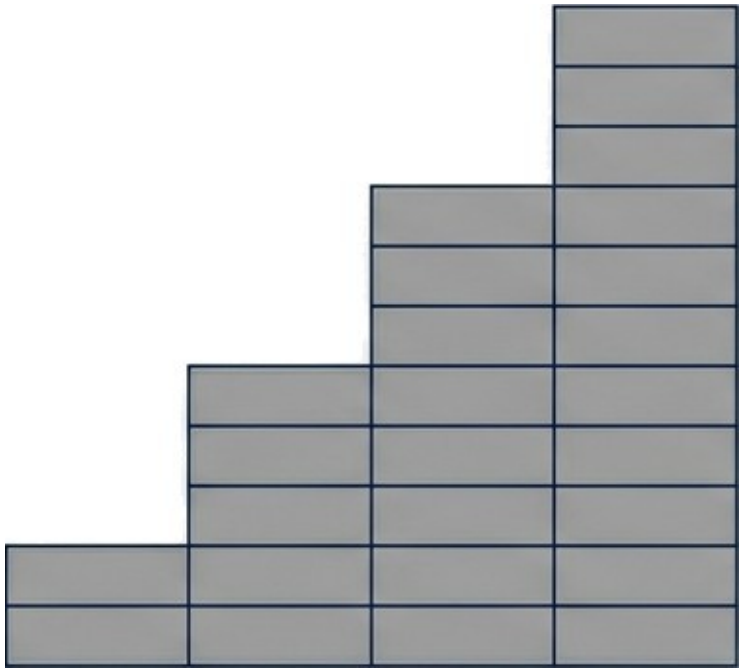
Trade-off: Sacrificed strict structure for **scalability & flexibility**

The modern frontier: Optimization & Intelligence



AI-native: Learned indexes

Traditional index **Learned index**



From sorting to predicting



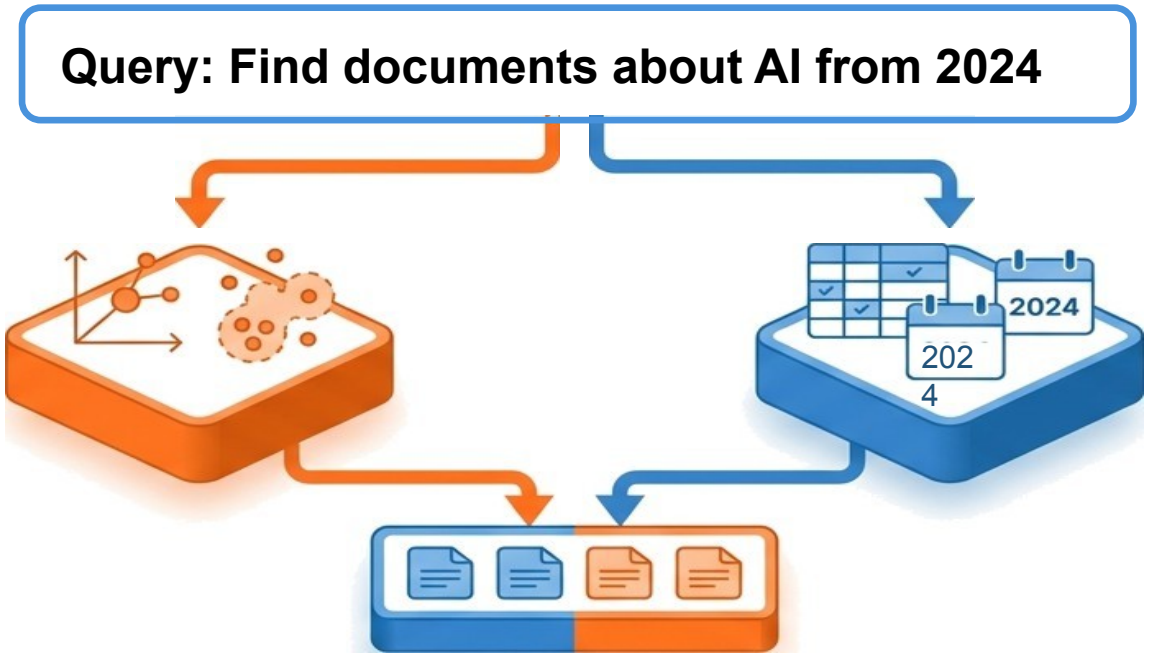
Sorting data into bins

Predicting data position via CDF

Tim Kraska et al. (2018). **The Case for Learned Index Structures**. In Proceedings of the 2018 International Conference on Management of Data (SIGMOD '18).

Vector & Hybrid search

Vector Search
(Finding Meaning/Concepts)



Relational Filter
(Finding Facts/Dates)

Result List

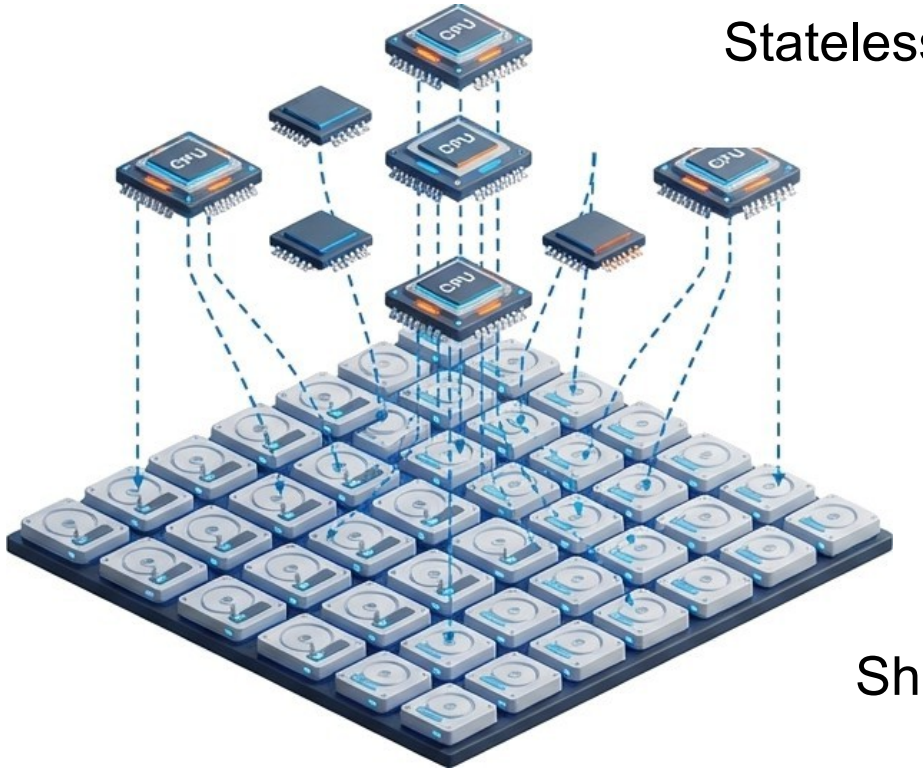
Problem: Vectors understand meaning but miss hard facts.
Solution: Hybrid Engines (VBASE).

Zhang et al. (2023). **VBASE: Unifying Online Vector Similarity Search and Relational Queries via Relaxed Monotonicity**. 17th USENIX Symposium on Operating Systems Design and Implementation (OSDI '23).

Cloud-native: Infinite elasticity

Scale-to-Zero: Pay only for what you use.

Elasticity: Spin up thousands of nodes in milliseconds.



Stateless compute nodes

Shared storage pool (S3/Blob)

Verbitski et al. (2017) **Amazon Aurora: Design Considerations for High Throughput Cloud-Native Relational Databases** (SIGMOD '17).

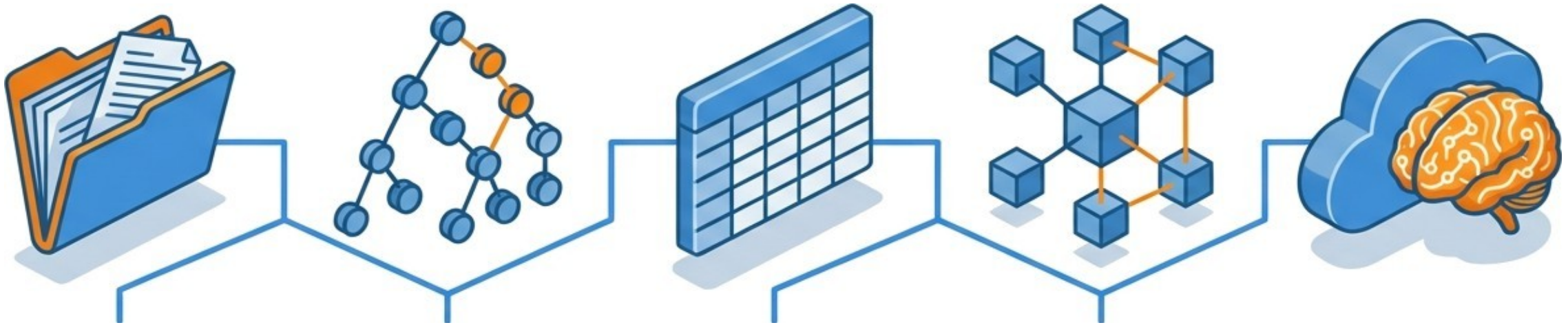
Zero-ETL & The Lakehouse



- **Best of both worlds:** Low-cost storage of a Lake + ACID transactions of a Warehouse.
- **Zero-ETL:** Query live data directly without complex pipelines.
- **Universal formats:** Apache Iceberg & AnyBlox.

Armbrust et al. (2020) **Delta Lake: High-Performance ACID Table Storage over Cloud Object Stores** (VLDB '20).

The evolution of data models



1960s

Chaos:

Unstructured file systems, lack of standardization.

1970s

Complexity:

Hierarchical and network models, rigid structures.

1980s

Logic/SQL:

Relational models, structured query language, ACID

2010s

NoSQL/Scale:

Document, graph, key-value stores, horizontal scaling.

Today

Intelligence:

AI-powered models, vector databases, hybrid systems, autonomous data management.

Takeaway messages

- **The foundation:** Moving from chaotic files to structured tables.
- **The mechanics:** Understanding rows, columns, keys, and SQL.
- **The evolution:** From hierarchical trees to relational grids.
- **The future:** AI-driven, serverless, and unified Lakehouse.

Database literacy is the core skill for the next generation of IT professionals.

MCQs

MCQ 1 - Which statement best describes a database?

A. A collection of text files stored on a computer without a fixed structure or relationship

B. A spreadsheet created using Excel, mainly designed for simple calculations, small datasets, and single-user editing

C. A collection of structured data managed for efficient access and update

D. A programming language used to store data

MCQs

MCQ 2 - Which of the following correctly distinguishes a database from a DBMS?

A. A database is a software application, while a DBMS is the raw data stored on disk

B. A database stores data; a DBMS manages and accesses the data

C. A DBMS is used only for backup and recovery purposes and does not support querying, updating, security, or multi-user access

D. There is no difference between a database and a DBMS; the two terms refer to exactly the same concept and can be used interchangeably in all contexts

MCQs

MCQ 3 - In a database table, which term refers to one complete record of a student?

A. Column

B. Cell

C. Row

D. Database

MCQs

MCQ 4 - Which of the following is the best choice for a primary key in a STUDENTS table?

A. Student name

B. Birthday

C. Student ID

D. Major

MCQs

MCQ 5 - Which problem is best handled by a database but not by Excel?

A. Formatting tables with colours, borders, fonts for better visualization

B. Calculating averages, totals, and basic statistics from a set of data

C. Allowing many users to safely update data at the same time

D. Printing large tables with customized page layouts, headers, and footers